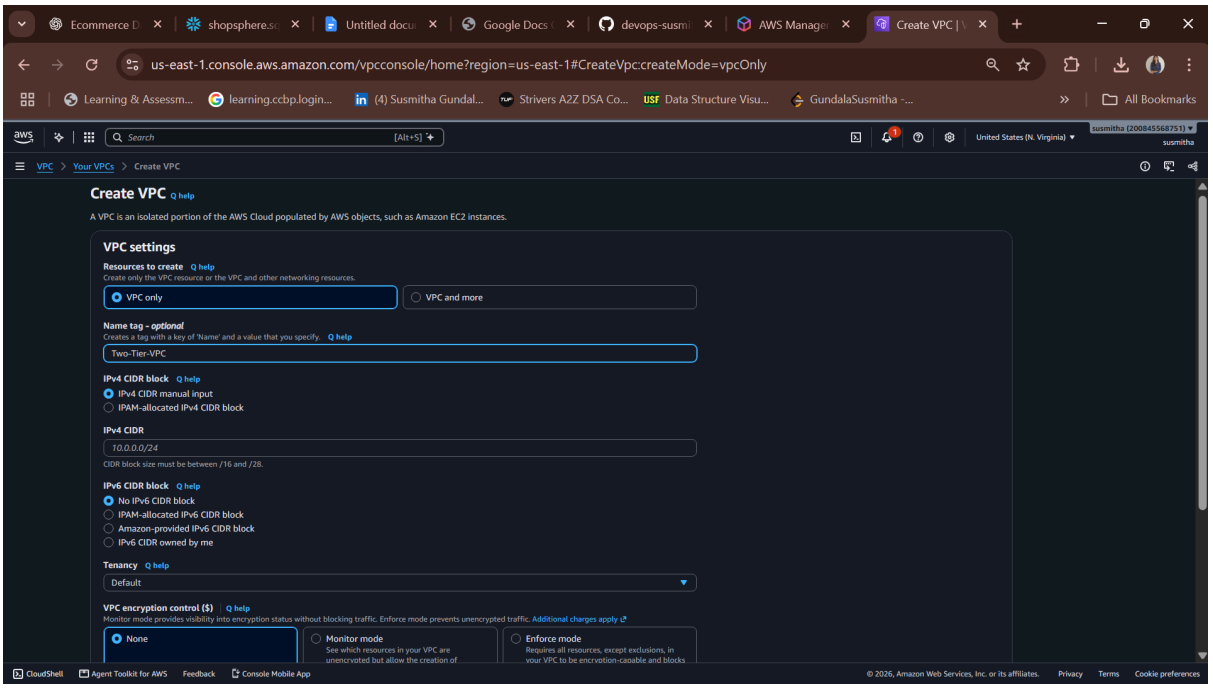
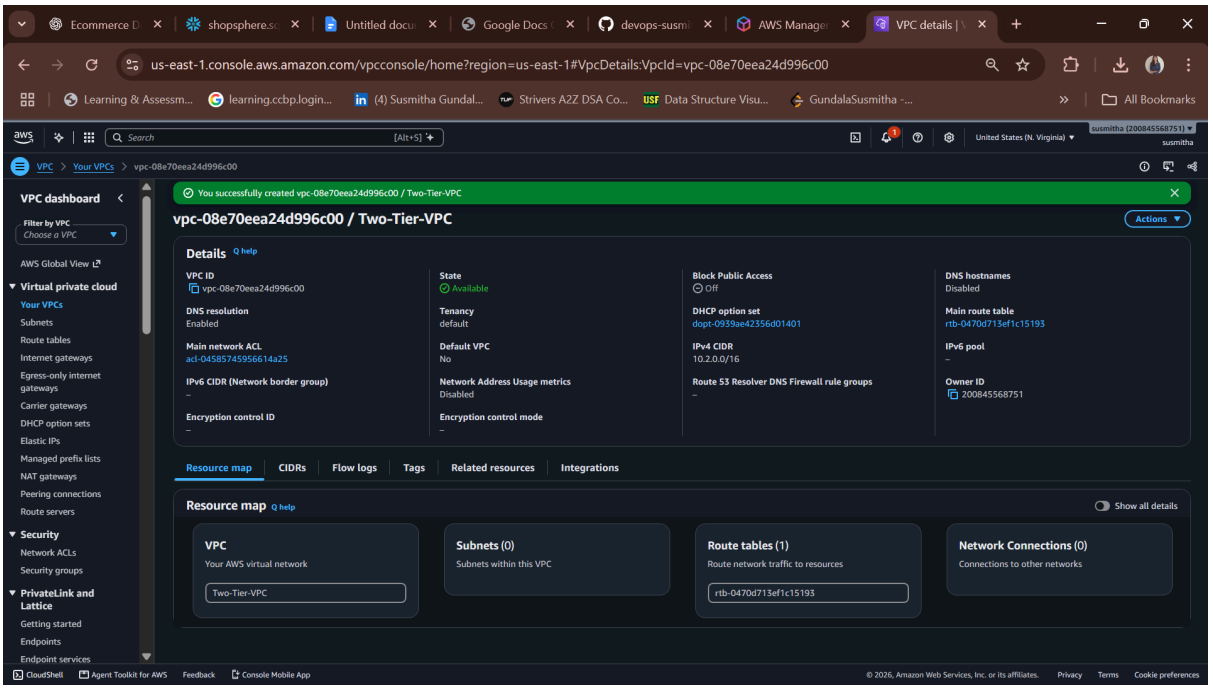
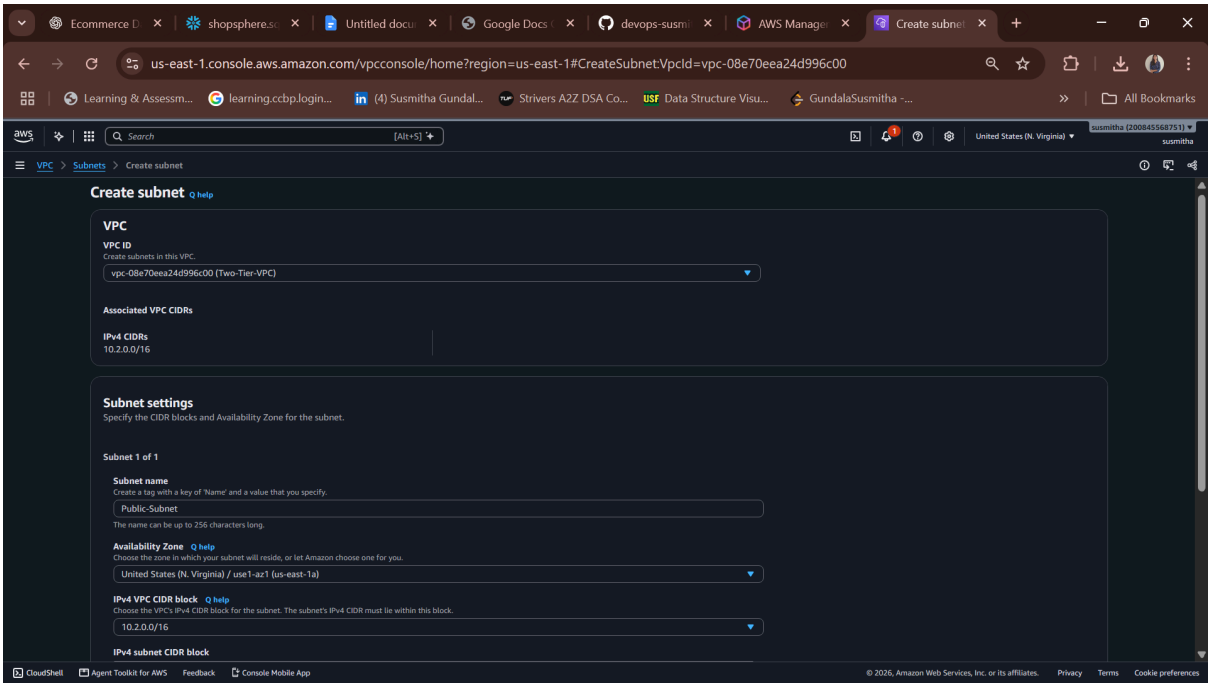
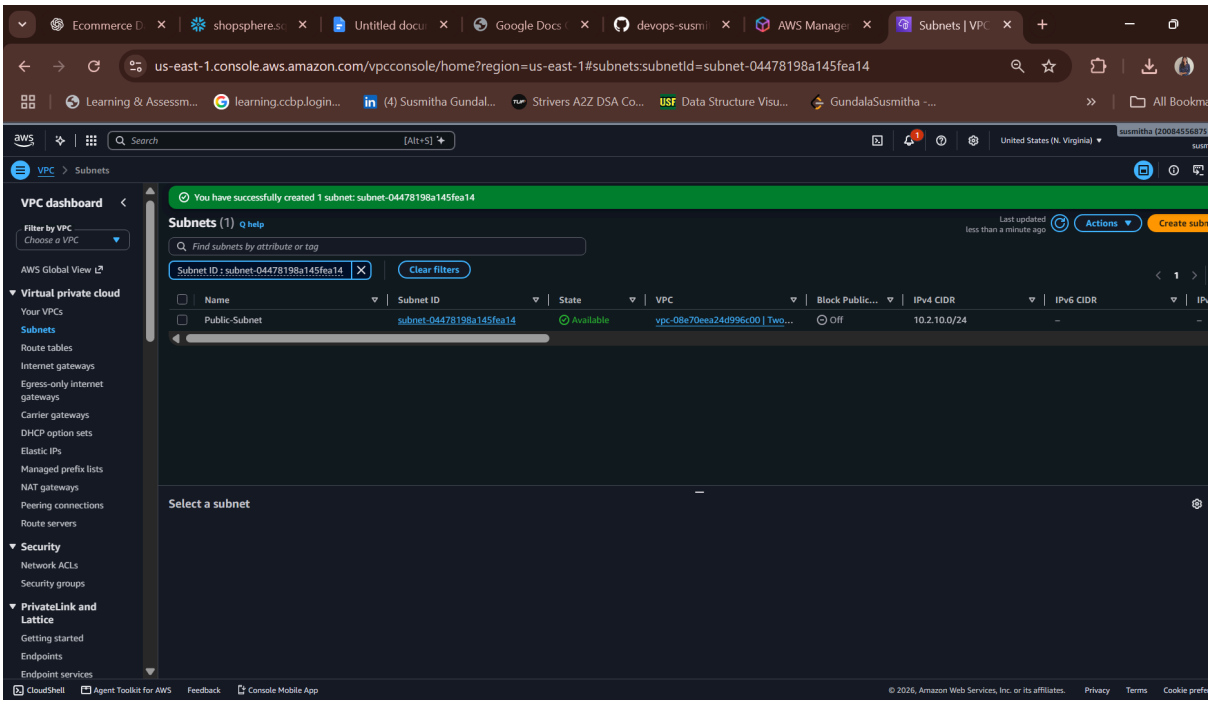
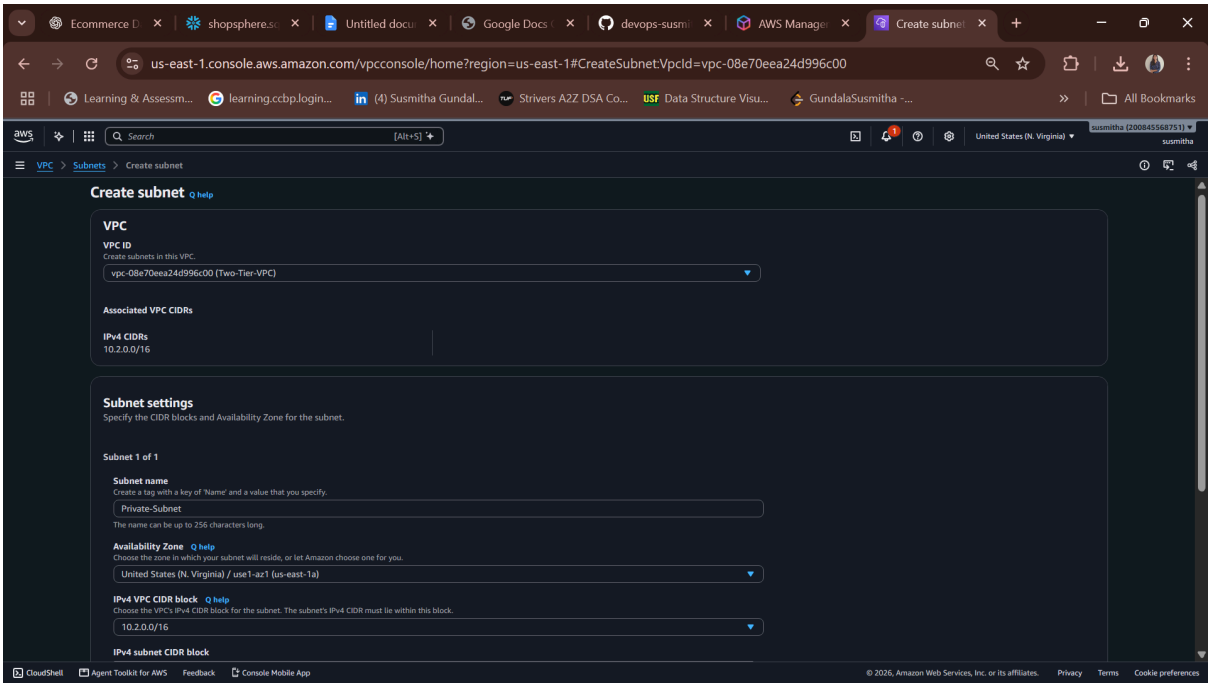
AWS VPC Networking and Two-Tier Web Application Deployment

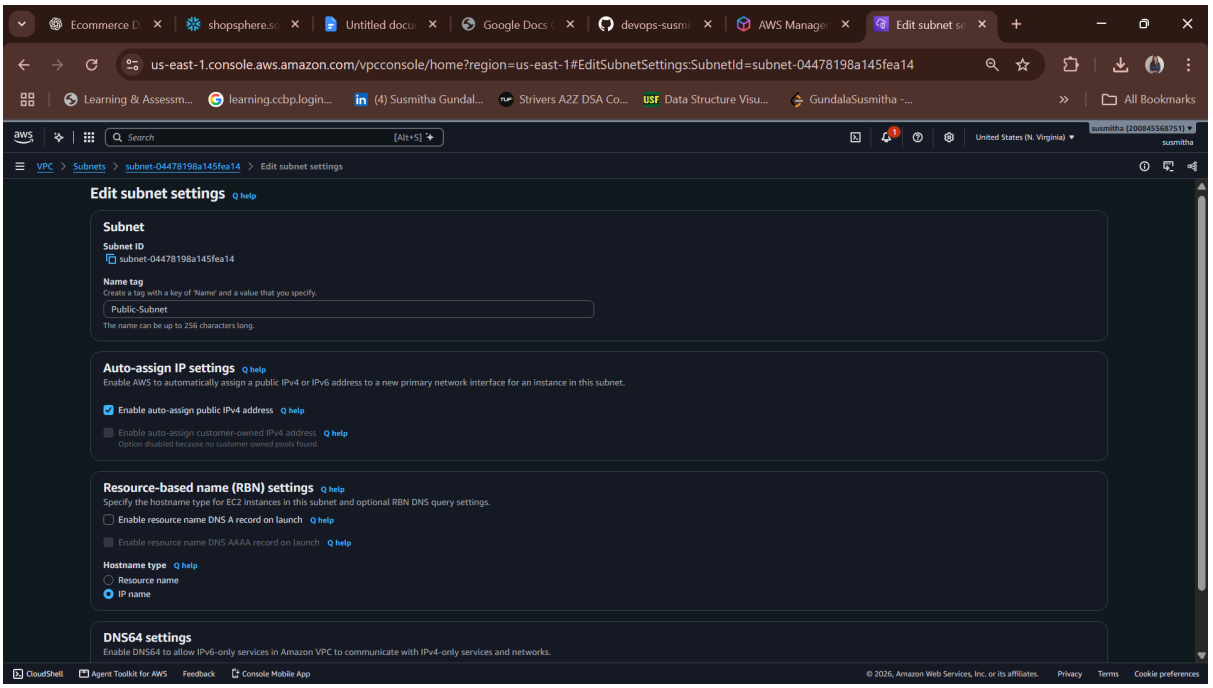


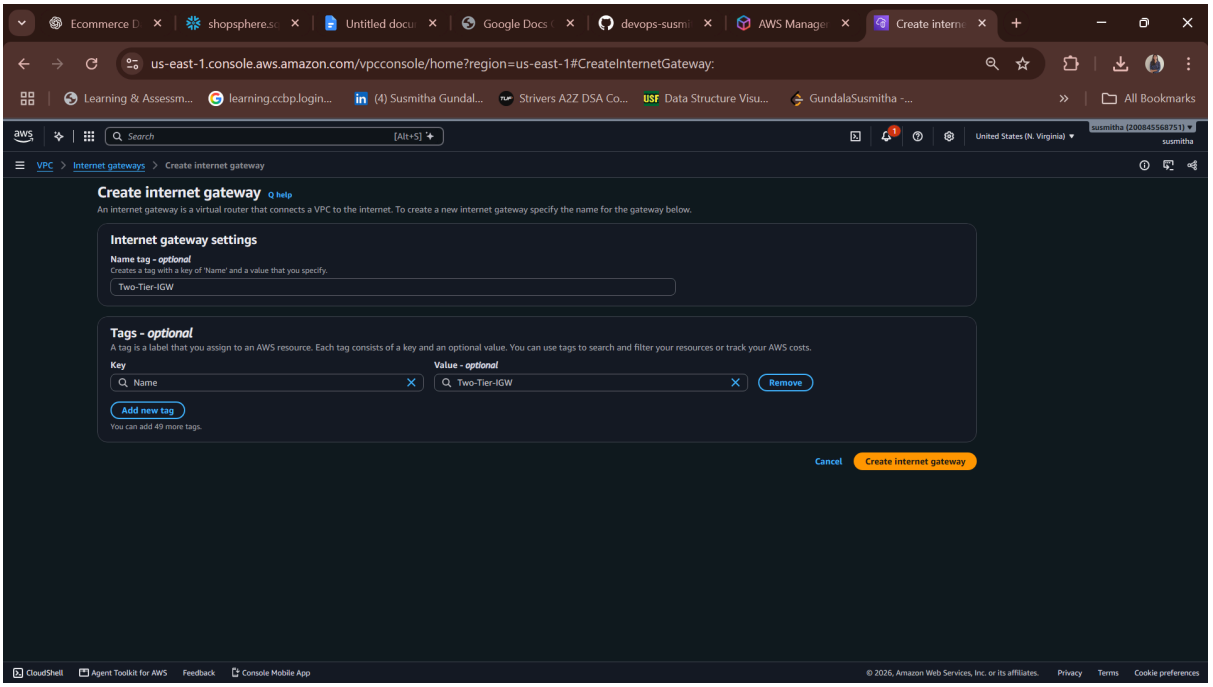


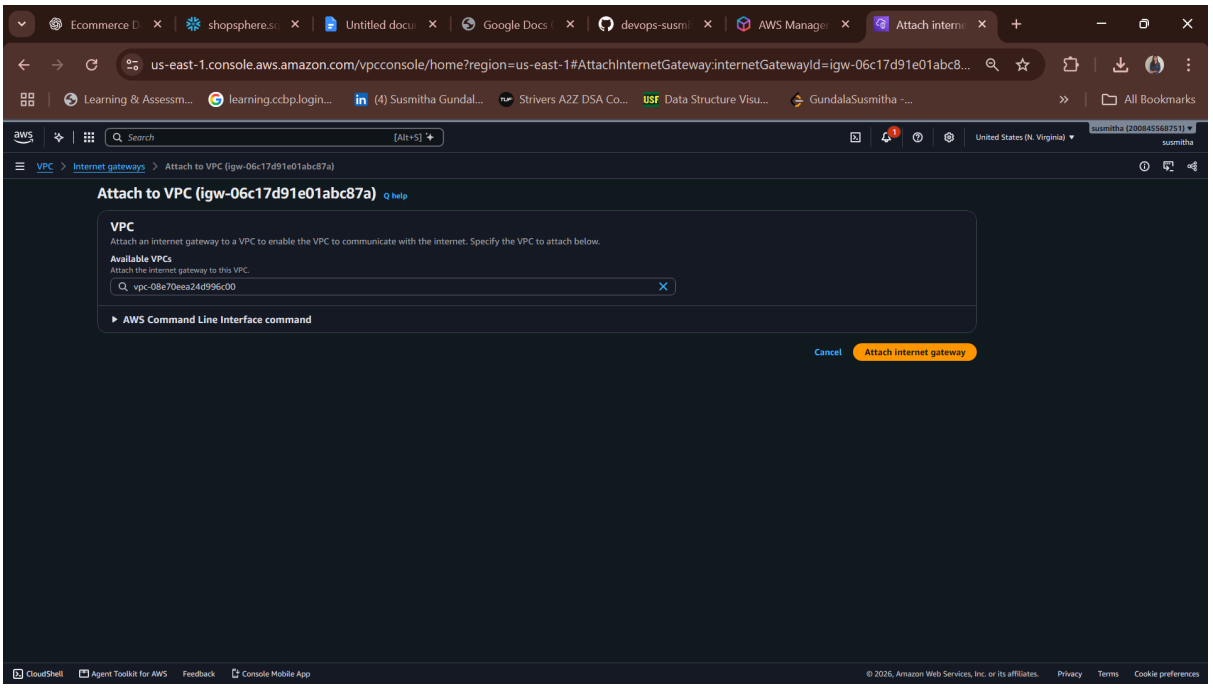

Create subnet
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateSubnet:VpcId=vpc-08e70eea24d996c00
VPC > Subnets > Create subnet
Create subnet
VPC
VPC ID
Create subnets in this VPC.
vpc-08e70eea24d996c00 (Two-Tier-VPC)
Associated VPC CIDRs
IPv4 CIDRs
10.2.0.0/16
Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-Subnet
The name can be up to 256 characters long.
Availability Zone
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1a)
IPv4 VPC CIDR block
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.2.0.0/16
IPv4 subnet CIDR block

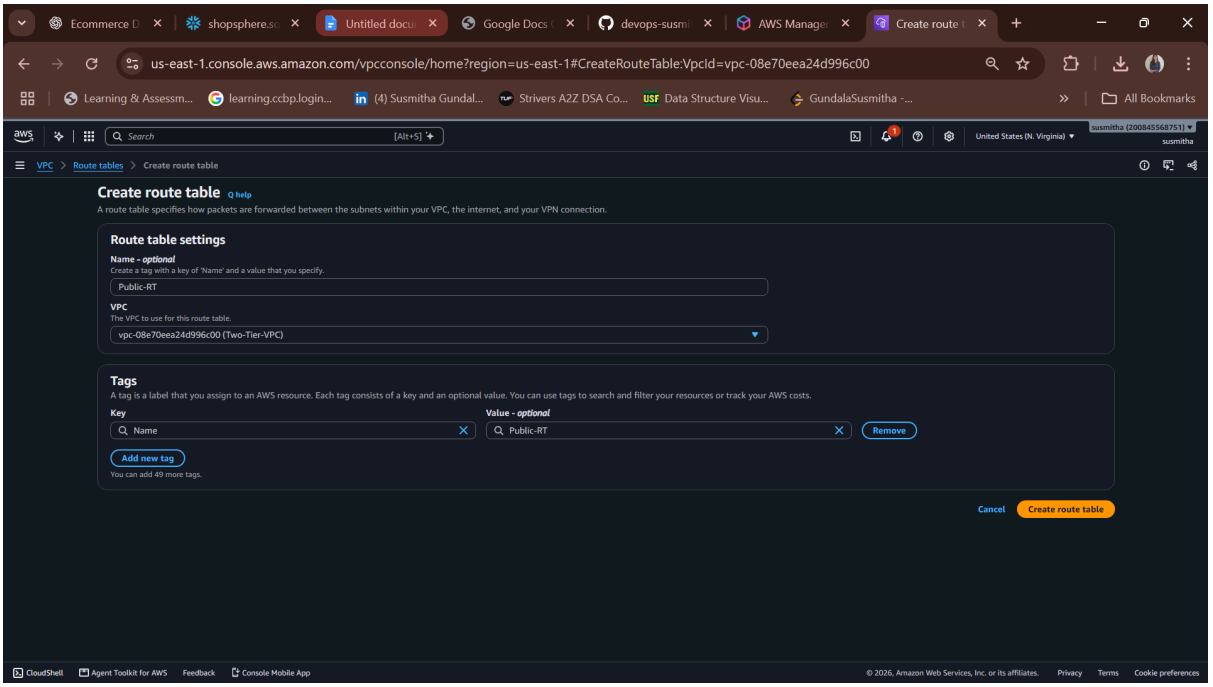

Subnets | VPC
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:subnetId=subnet-04478198a145fea14
VPC > Subnets
VPC dashboard
You have successfully created 1 subnet: subnet-04478198a145fea14
Subnets (1)
Find subnets by attribute or tag
Subnet ID : subnet-04478198a145fea14
Clear filters
Actions
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
IPv6 CIDR
Public-Subnet
subnet-04478198a145fea14
Available
vpc-08e70eea24d996c00 | Two...
Off
10.2.10.0/24
Select a subnet
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services

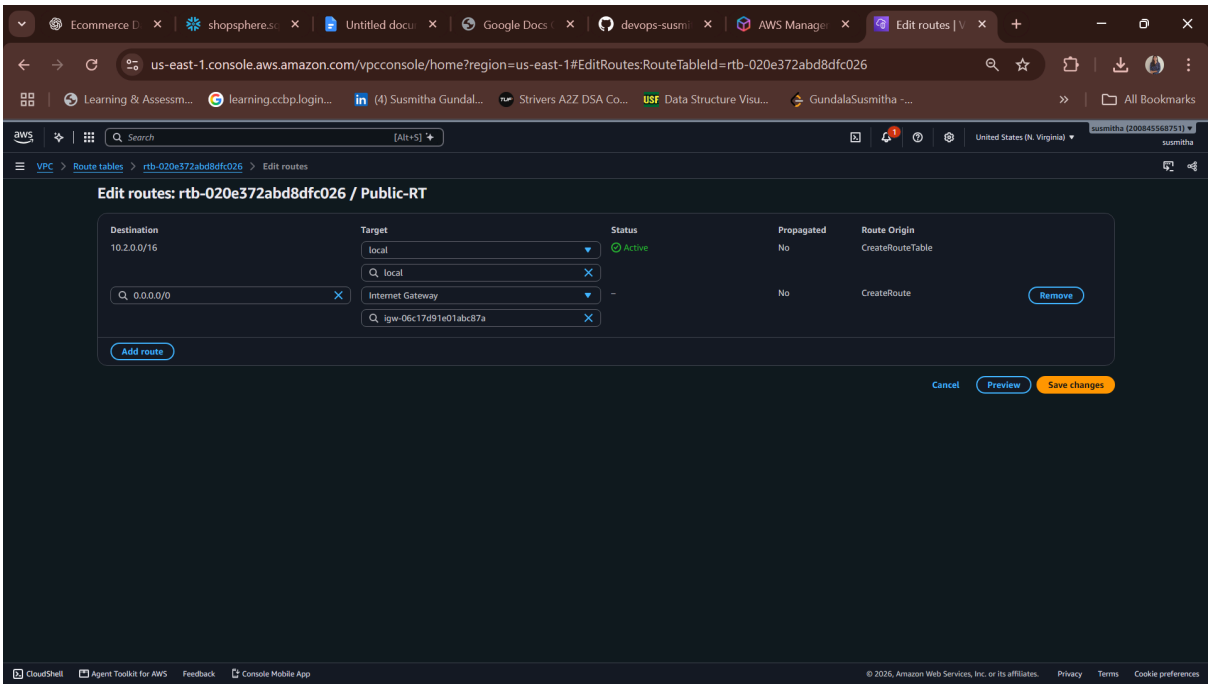

Create subnet
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateSubnet:VpcId=vpc-08e70eea24d996c00
VPC > Subnets > Create subnet
Create subnet
VPC
VPC ID
Create subnets in this VPC.
vpc-08e70eea24d996c00 (Two-Tier-VPC)
Associated VPC CIDRs
IPv4 CIDRs
10.2.0.0/16
Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.
Subnet 1 of 1
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private-Subnet
The name can be up to 256 characters long.
Availability Zone
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1a)
IPv4 VPC CIDR block
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.2.0.0/16
IPv4 subnet CIDR block

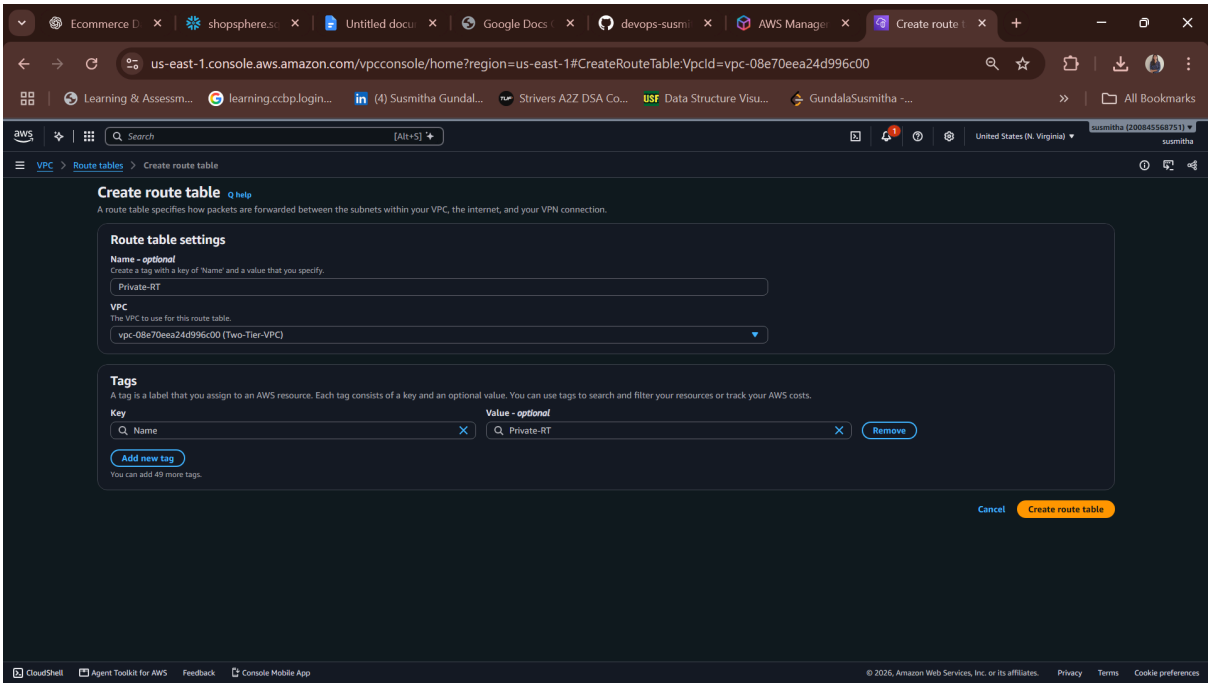

Edit subnet settings
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditSubnetSettings:SubnetId=subnet-04478198a145fea14
VPC > Subnets > subnet-04478198a145fea14 > Edit subnet settings
Edit subnet settings
Subnet
Subnet ID
subnet-04478198a145fea14
Name tag
Create a tag with a key of 'Name' and a value that you specify.
Public-Subnet
The name can be up to 256 characters long.
Auto-assign IP settings
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.
Enable auto-assign public IPv4 address
Enable auto-assign customer-owned IPv4 address
Option disabled because no customer owned pools found.
Resource-based name (RBN) settings
Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.
Enable resource name DNS A record on launch
Enable resource name DNS AAAA record on launch
Hostname type
Resource name
IP name
DNS64 settings
Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

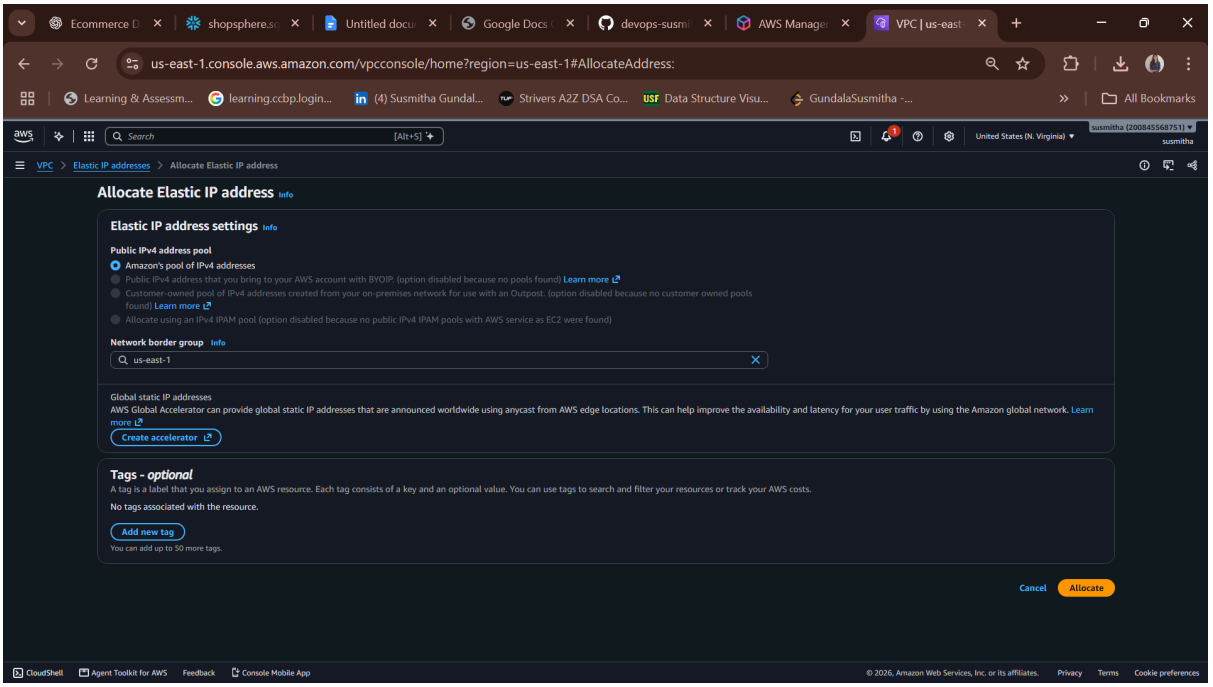

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateInternetGateway:
VPC > Internet gateways > Create internet gateway
Create internet gateway
An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.
Internet gateway settings
Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.
Two-Tier-IGW
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
Two-Tier-IGW
Remove
Add new tag
You can add 49 more tags.
Cancel
Create internet gateway

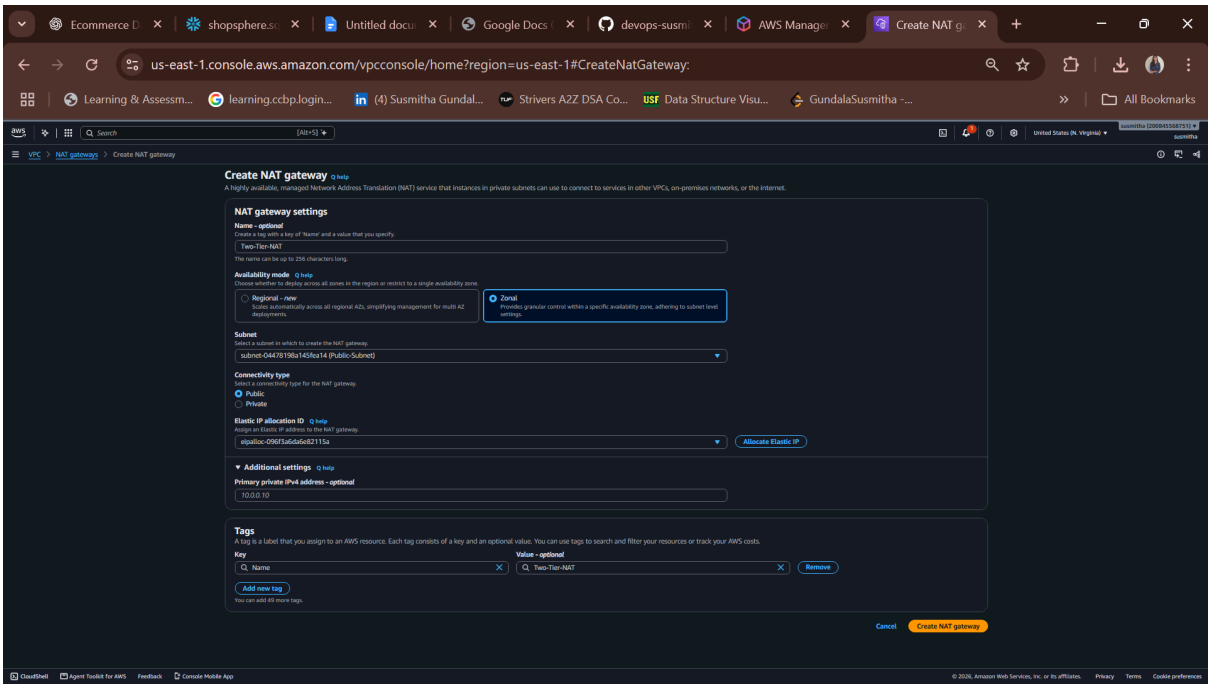

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#AttachInternetGateway:internetGatewayId=igw-06c17d91e01abc8...
VPC > Internet gateways > Attach to VPC (igw-06c17d91e01abc87a)
Attach to VPC (igw-06c17d91e01abc87a)
VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.
Available VPCs
Attach the internet gateway to this VPC.
vpc-08e70eea24d996c00
AWS Command Line Interface command
Cancel
Attach internet gateway

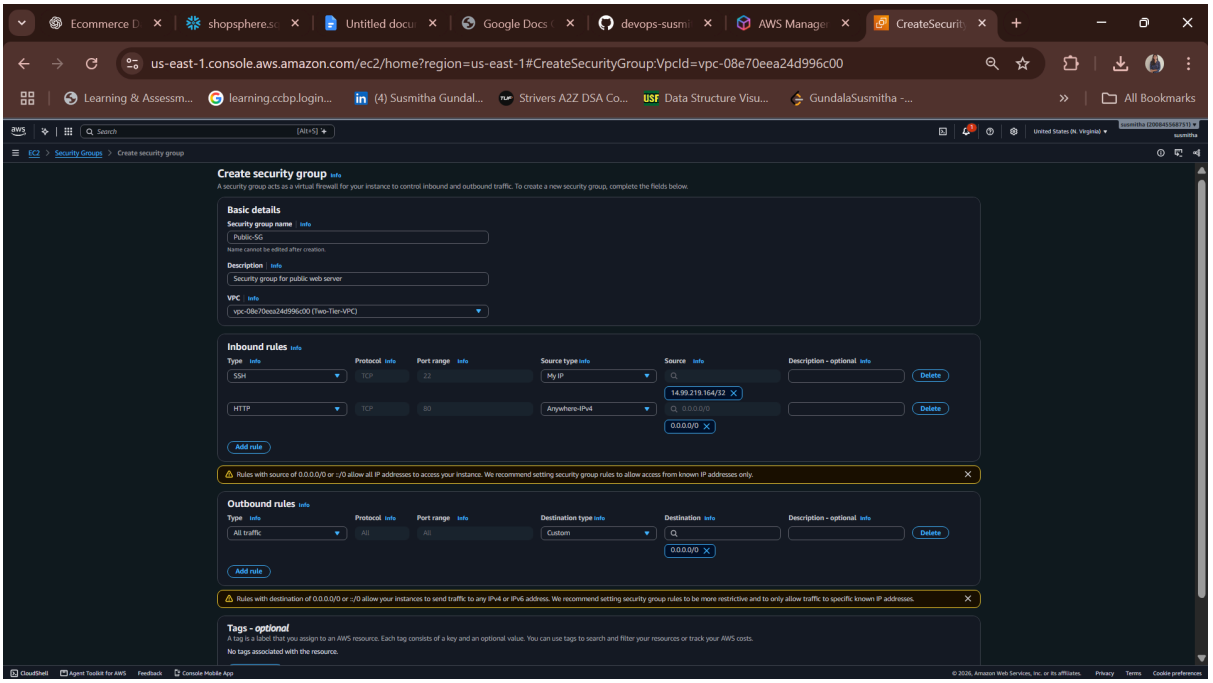

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateRouteTable:VpcId=vpc-08e70eea24d996c00
VPC > Route tables > Create route table
Create route table
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Public-RT
VPC
The VPC to use for this route table.
vpc-08e70eea24d996c00 (Two-Tier-VPC)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
Public-RT
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table

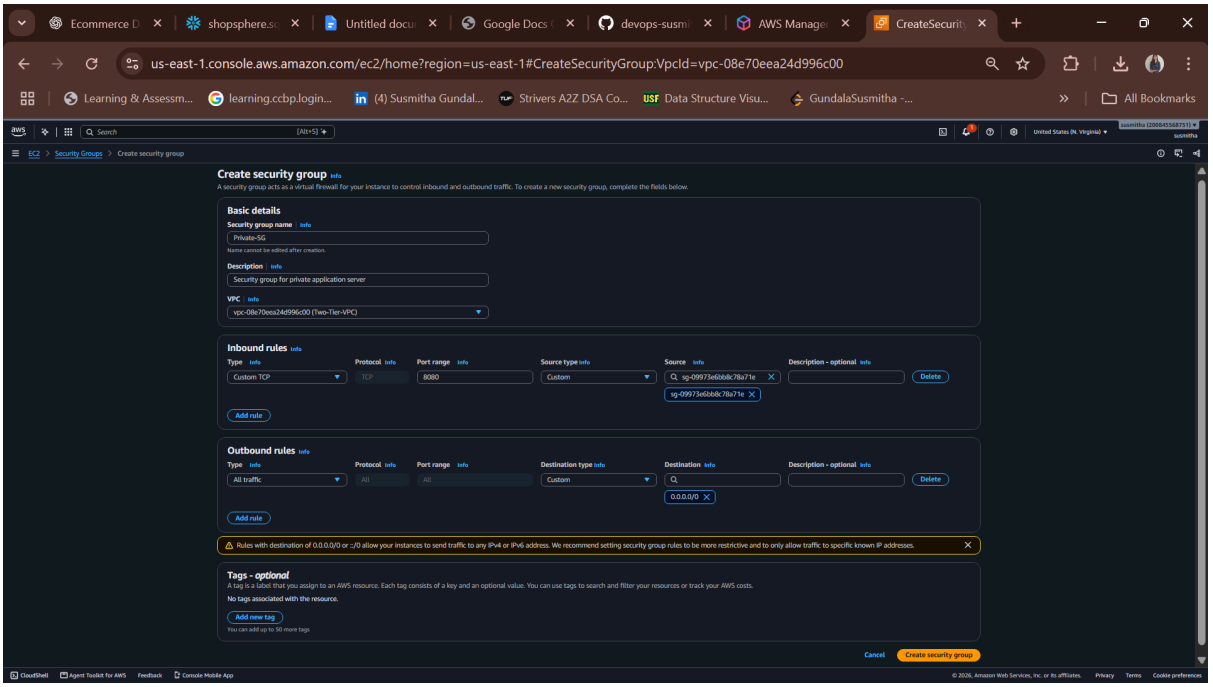

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRoutes:RouteTableId=rtb-020e372abd8dfc026
VPC > Route tables > rtb-020e372abd8dfc026 > Edit routes
Edit routes: rtb-020e372abd8dfc026 / Public-RT
Destination
Target
Status
Propagated
Route Origin
10.2.0.0/16
local
Active
No
CreateRouteTable
local
0.0.0.0/0
Internet Gateway
No
CreateRoute
Remove
igw-06c17d91e01abc87a
Add route
Cancel
Preview
Save changes

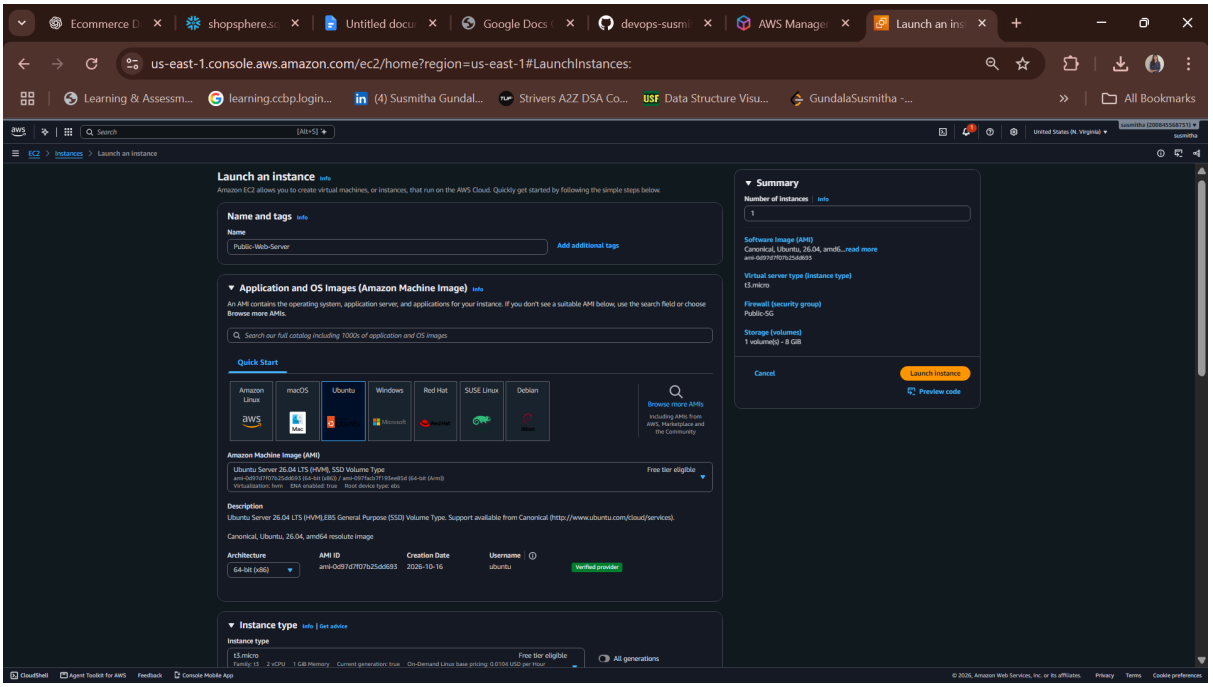

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateRouteTable:VpcId=vpc-08e70eea24d996c00
VPC > Route tables > Create route table
Create route table
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Private-RT
VPC
The VPC to use for this route table.
vpc-08e70eea24d996c00 (Two-Tier-VPC)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
Private-RT
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

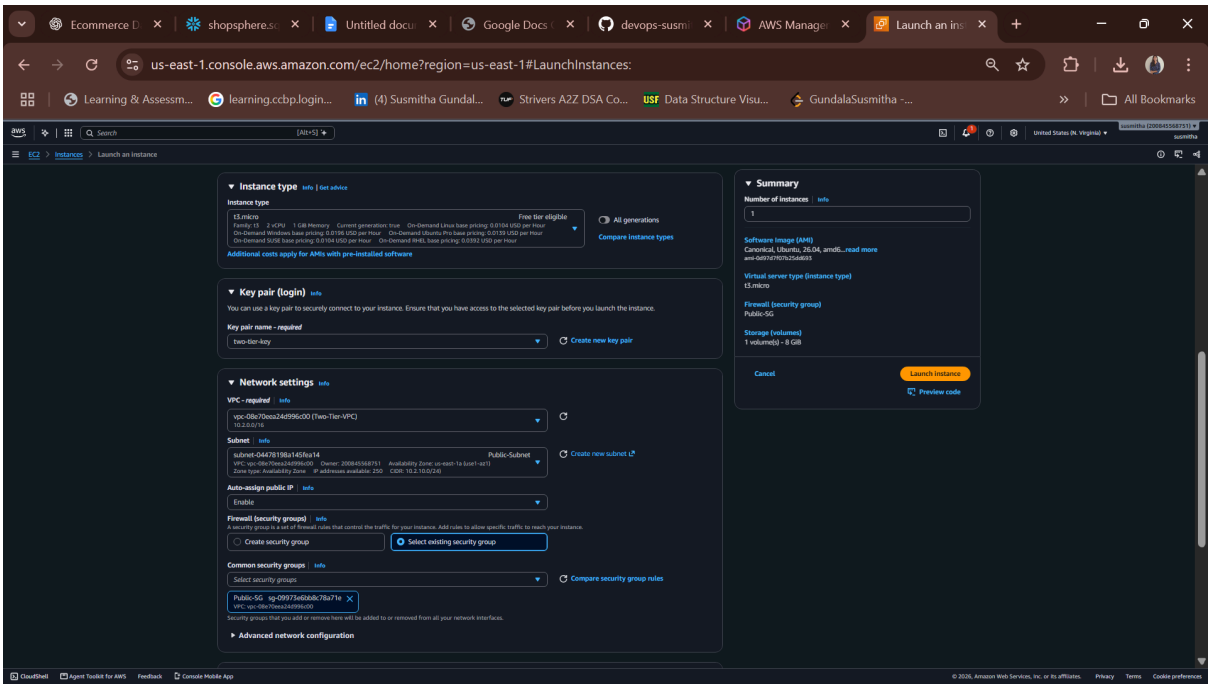

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#AllocateAddress:
VPC > Elastic IP addresses > Allocate Elastic IP address
Allocate Elastic IP address Info
Elastic IP address settings Info
Public IPv4 address pool
Amazon's pool of IPv4 addresses
Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) Learn more
Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) Learn more
Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)
Network border group Info
us-east-1
Global static IP addresses
AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. Learn more
Create accelerator
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Allocate

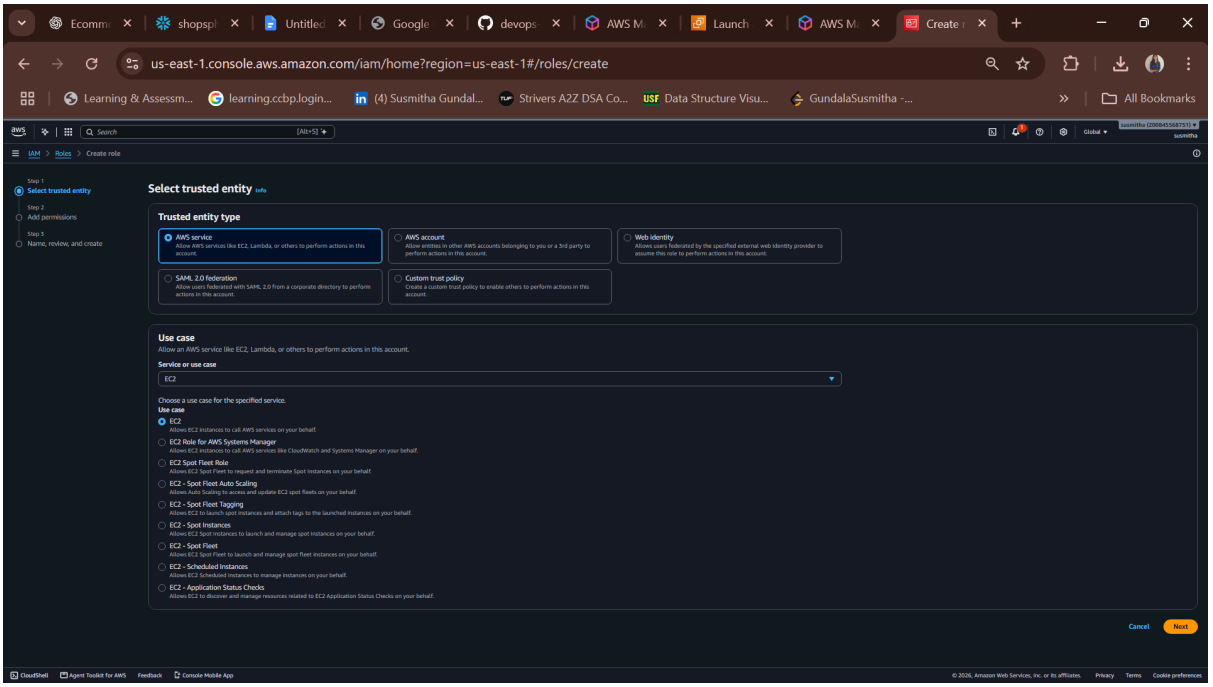

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateNatGateway:
VPC > NAT gateways > Create NAT gateway
Create NAT gateway Info
A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.
NAT gateway settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Two-Tier-NAT
The name can be up to 256 characters long.
Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.
Regional - new
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.
Zonal
Provides granular control within a specific availability zone, adhering to subnet level settings.
Subnet
Select a subnet in which to create the NAT gateway.
subnet-04478198a145fea14 (Public-Subnet)
Connectivity type
Select a connectivity type for the NAT gateway.
Public
Private
Elastic IP allocation ID Info
Assign an Elastic IP address to the NAT gateway.
eipalloc-096f3a6da6e82115a
Allocate Elastic IP
Additional settings Info
Primary private IPv4 address - optional
10.0.0.10
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Value - optional
Name
Two-Tier-NAT
Remove
Add new tag
You can add 49 more tags.
Cancel
Create NAT gateway

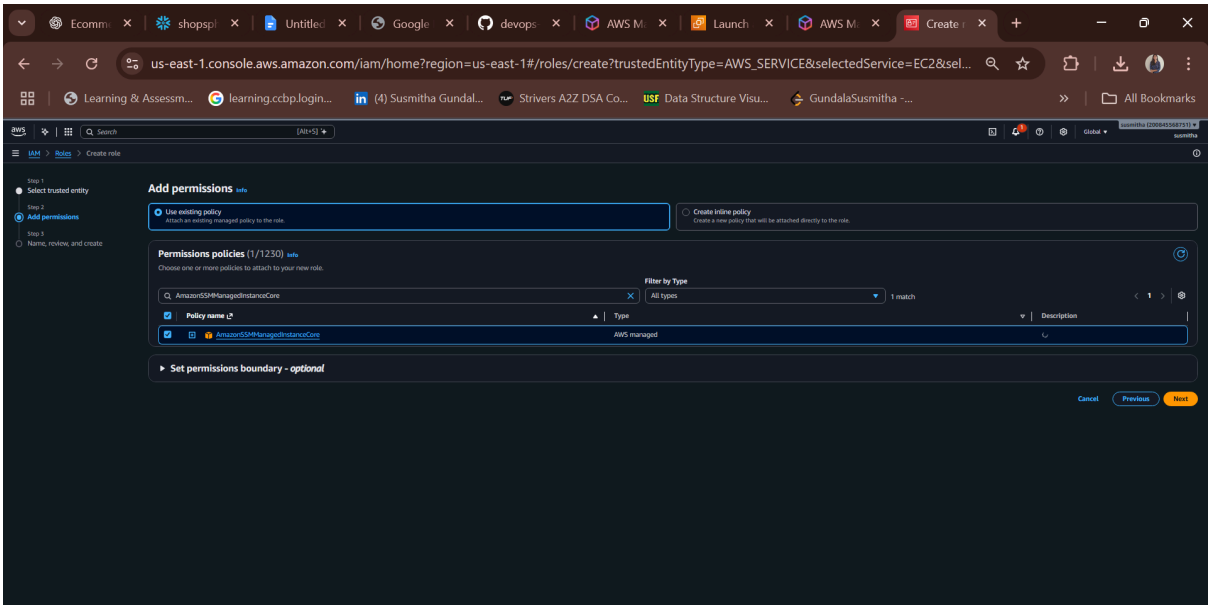

Ecommerce D
shopsphere.sc
Untitled docu
Google Docs
devops-susmi
AWS Manage
CreateSecurit
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSecurityGroup:VpcId=vpc-08e70eea24d996c00
Learning & Assessm...
learning.ccbp.login...
(4) Susmitha Gundal...
Strivers A2Z DSA Co...
Data Structure Visu...
GundalaSusmitha -...
All Bookmarks
EC2 > Security Groups > Create security group
Create security group
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.
Basic details
Security group name
Public-SG
Name cannot be edited after creation.
Description
Security group for public web server
VPC
vpc-08e70eea24d996c00 (Two-Tier-VPC)
Inbound rules
Type
Protocol
Port range
Source type
Source
Description - optional
SSH
TCP
22
My IP
14.99.219.164/32
Delete
HTTP
TCP
80
Anywhere-IPv4
0.0.0.0/0
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Outbound rules
All traffic
All
Destination type
Destination
Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences

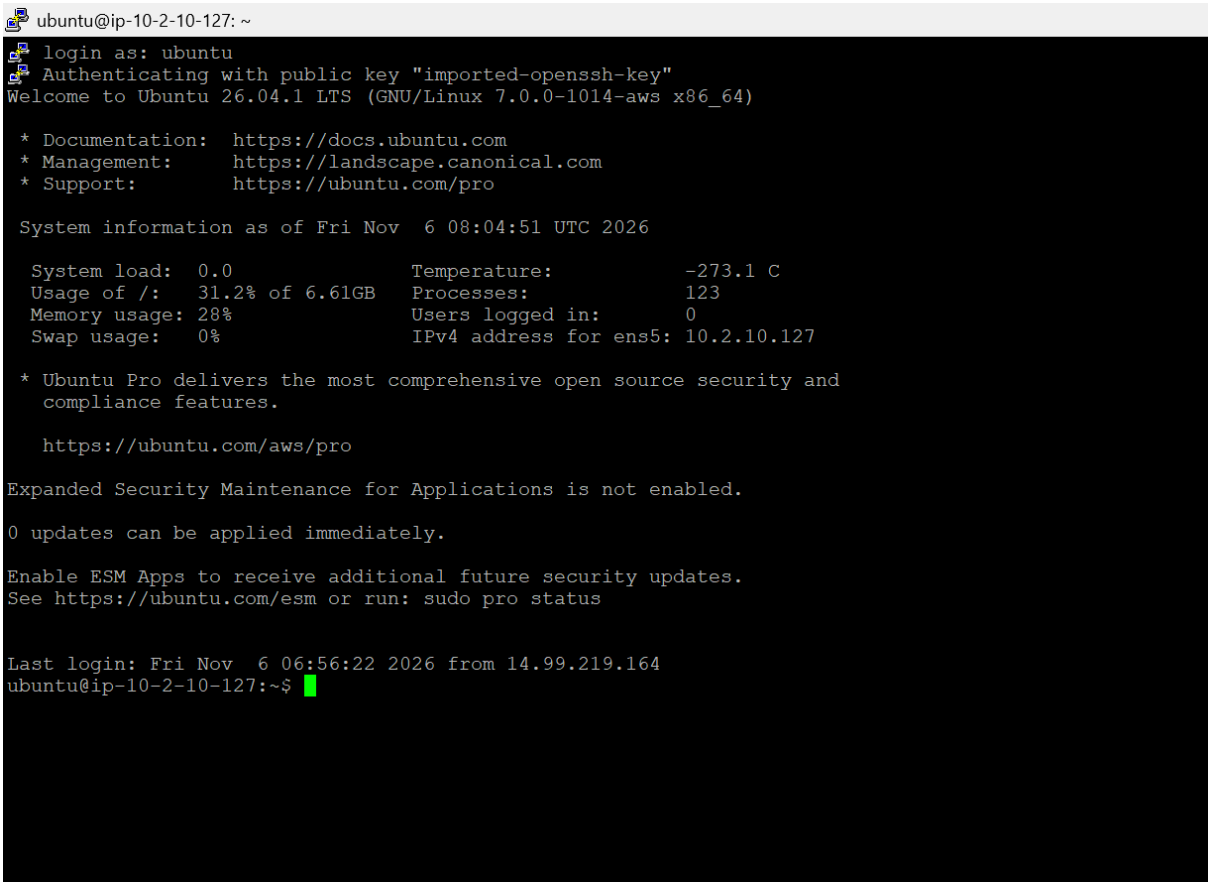

Ecommerce D
shopsphere.sc
Untitled docu
Google Docs
devops-susmi
AWS Manage
CreateSecurit
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSecurityGroup:VpcId=vpc-08e70eea24d996c00
Learning & Assessm...
learning.ccbp.login...
(4) Susmitha Gundal...
Strivers A2Z DSA Co...
Data Structure Visu...
GundalaSusmitha -...
All Bookmarks
EC2 > Security Groups > Create security group
Create security group
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.
Basic details
Security group name
Private-SG
Name cannot be edited after creation.
Description
Security group for private application server
VPC
vpc-08e70eea24d996c00 (Two-Tier-VPC)
Inbound rules
Type
Protocol
Port range
Source type
Source
Description - optional
Custom TCP
TCP
8080
Custom
sg-09973e6bb8c78a71e
Delete
Add rule
Outbound rules
All traffic
All
Destination type
Destination
Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.
Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Create security group
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences

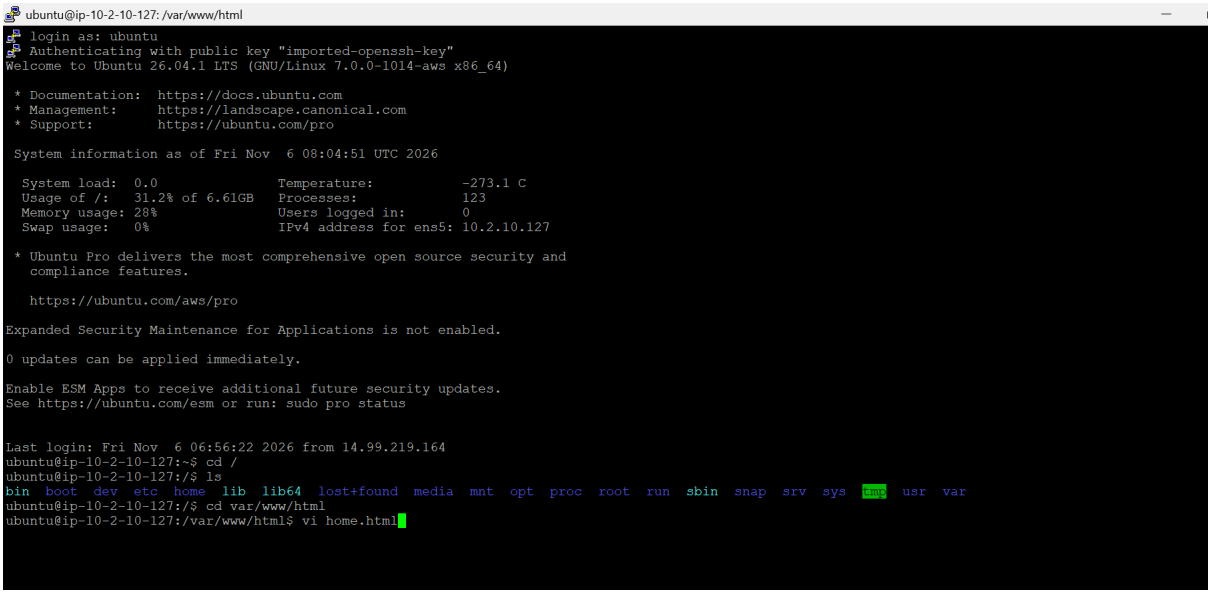

Launch an instance
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags
Name
Public-Web-Server
Add additional tags
Application and OS Images (Amazon Machine Image)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Ubuntu Server 26.04 LTS (HVM), SSD Volume Type
Free tier eligible
Description
Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).
Canonical, Ubuntu, 26.04, amd64 resolute image
Architecture
64-bit (x86)
AMI ID
ami-0d97d7f07b25d6935
Creation Date
2026-10-16
Username
ubuntu
Verified provider
Instance type
Instance type
t3.micro
Free tier eligible
All generations
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
Public-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code


Instance type
Instance type
t3.micro
Free tier eligible
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login)
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
two-tier-key
Create new key pair
Network settings
VPC - required
vpc-08e70eca24d996c00 (Two-Tier-VPC)
Subnet
subnet-04478198a145fea14
Public-Subnet
Create new subnet
Auto-assign public IP
Enable
Firewall (security groups)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups
Select security groups
Compare security group rules
Public-SG sg-09971e6bb8c78a71e
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 26.04, amd6...read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
Public-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

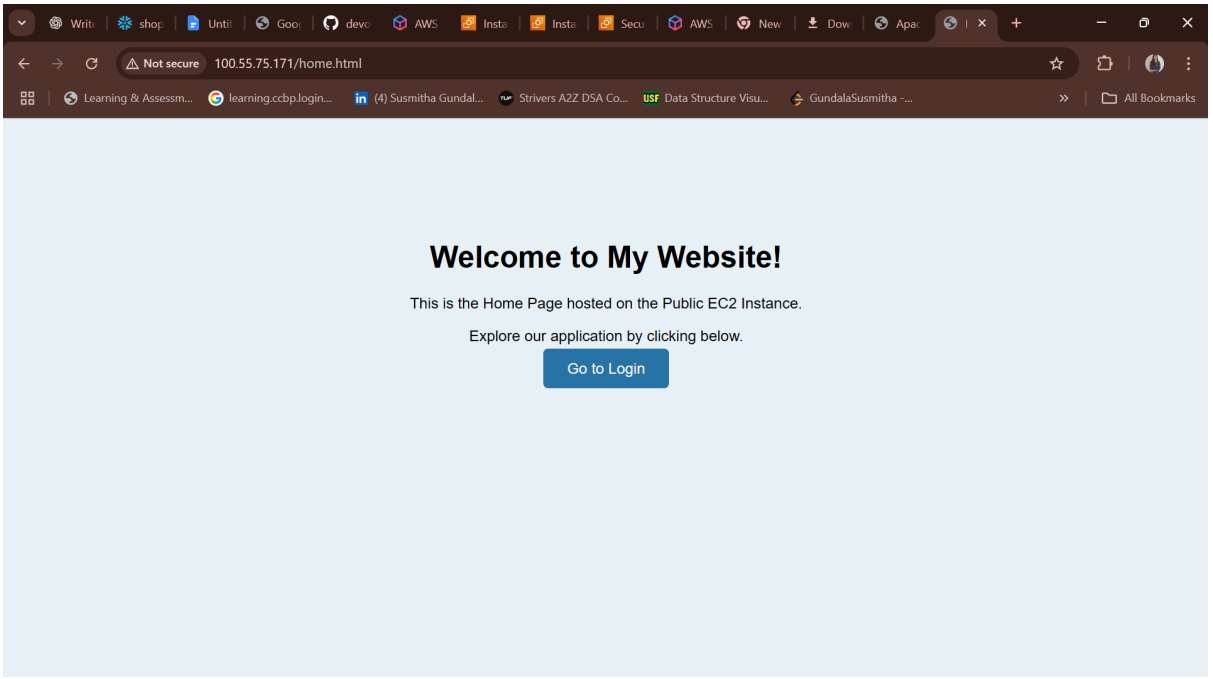

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create
Select trusted entity
Trusted entity type
AWS service
AWS account
Web identity
SAML 2.0 federation
Custom trust policy
Use case
Service or use case
EC2
Cancel
Next

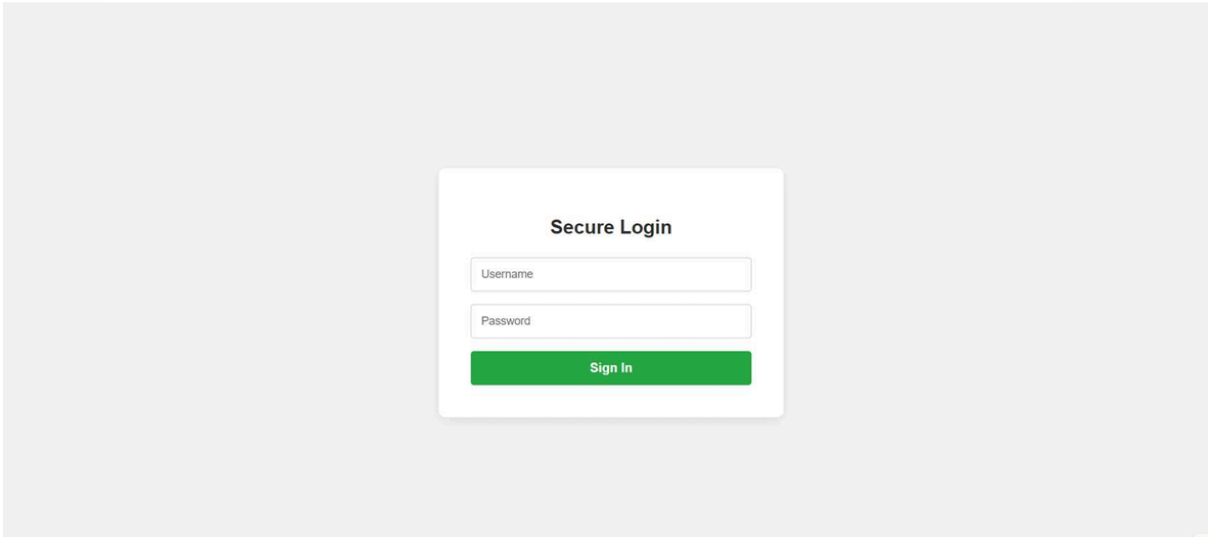

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=EC2&sel...
Add permissions
Use existing policy
Create inline policy
Permissions policies (1/1230)
AmazonSSMManagedInstanceCore
AWS managed
Set permissions boundary - optional
Cancel
Previous
Next


ubuntu@ip-10-2-10-127: ~
login as: ubuntu
Authenticating with public key "imported-openssh-key"
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Fri Nov 6 08:04:51 UTC 2026
System load: 0.0
Usage of /: 31.2% of 6.61GB
Memory usage: 28%
Swap usage: 0%
Temperature: -273.1 C
Processes: 123
Users logged in: 0
IPv4 address for ens5: 10.2.10.127
* Ubuntu Pro delivers the most comprehensive open source security and compliance features.
https://ubuntu.com/aws/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Fri Nov 6 06:56:22 2026 from 14.99.219.164
ubuntu@ip-10-2-10-127:~$


ubuntu@ip-10-2-10-127: /var/www/html
login as: ubuntu
Authenticating with public key "imported-openssh-key"
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Fri Nov 6 08:04:51 UTC 2026
System load: 0.0
Usage of /: 31.2% of 6.61GB
Memory usage: 28%
Swap usage: 0%
Temperature: -273.1 C
Processes: 123
Users logged in: 0
IPv4 address for ens5: 10.2.10.127
* Ubuntu Pro delivers the most comprehensive open source security and compliance features.
https://ubuntu.com/aws/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Fri Nov 6 06:56:22 2026 from 14.99.219.164
ubuntu@ip-10-2-10-127:~$ cd /
ubuntu@ip-10-2-10-127:/$ ls
bin boot dev etc home lib lib64 lost+found media mnt opt proc root run sbin snap srv sys tmp usr var
ubuntu@ip-10-2-10-127:/$ cd var/www/html
ubuntu@ip-10-2-10-127:/var/www/html$ vi home.html


Not secure
100.55.75.171
Learning & Assessm...
learning.ccbp.login...
(4) Susmitha Gundal...
Strivers A2Z DSA Co...
Data Structure Visu...
GundalaSusmitha -...
All Bookmarks
Apache2 Default Page
Ubuntu
It works!
This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should replace this file (located at /var/www/html/index.html) before continuing to operate your HTTP server.
If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.
Configuration Overview
Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed on this server.
The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:
/etc/apache2/
|-- apache2.conf
| `-- ports.conf
|-- mods-enabled


Not secure
100.55.75.171/home.html
Learning & Assessm...
learning.ccbp.login...
(4) Susmitha Gundal...
Strivers A2Z DSA Co...
Data Structure Visu...
GundalaSusmitha -...
All Bookmarks
Welcome to My Website!
This is the Home Page hosted on the Public EC2 Instance.
Explore our application by clicking below.
Go to Login


Secure Login
Username
Password
Sign In